



The National Egyptian E-learning University, Cairo, Egypt

Faculty of Computers and Information Technology

Department of Information Technology

NileGuide: AI-Powered Tourism Planner and Activity Management Platform

A Project Submitted to the Department of Information Technology, Faculty of Computers and Information Technology,

The National Egyptian E-learning University, Cairo, Egypt

In the Requirements of Graduation for the award of the degree of bachelor's in computer and information technology

BY

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Academic Year 2025-2026

ABSTRACT

NileGuide is an AI-powered tourism planning platform designed to help tourists discover activities in Egypt, organize

personal travel plans, manage wishlists, and receive assistance through a retrieval-augmented AI chatbot. The system

combines an Angular standalone frontend, an ASP.NET Core Web API backend, and an independent FastAPI RAG service. The

frontend provides a responsive web interface for tourists and administrators, including activity discovery, activity

details, wishlist, schedule, map, profile management, and dashboard screens. The backend manages authentication,

authorization, activities, cities, categories, users, profiles, reviews, reports, plans, wishlists, profile pictures,

and dashboard data using Entity Framework Core and SQL Server. The AI module allows authorized users to upload knowledge

files, extract and chunk their content, store metadata in MongoDB, store embeddings in Qdrant, and generate contextual

answers through a chat endpoint. The project aims to simplify tourism planning in Egypt while giving administrators a

structured way to manage content and improve the AI knowledge base without changing application code.

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List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
AI	Artificial Intelligence
JWT	JSON Web Token
RAG	Retrieval-Augmented Generation
SPA	Single Page Application
SSR	Server-Side Rendering
DTO	Data Transfer Object
ORM	Object-Relational Mapper
EF Core	Entity Framework Core
LLM	Large Language Model
UI	User Interface
UX	User Experience

Chapter 1 - Introduction

1.1 Introduction

Tourism platforms require clear information, reliable filtering, and simple planning tools. NileGuide addresses these needs by offering a web application for discovering tourism activities in Egypt and managing the user journey from exploration to planning. The application also supports administrators through a dashboard that manages users, activities, reports, and AI knowledge files.

1.2 Motivation

The motivation behind NileGuide is to reduce the effort required to plan tourism experiences. Tourists often need to compare cities, categories, prices, images, opening hours, providers, and activity details before making decisions. At the same time, administrators need a maintainable platform for updating activity data and improving support content. NileGuide combines these needs in one system.

1.3 Objectives

- Provide a responsive frontend for tourists to discover, filter, and review tourism activities.
- Allow authenticated tourists to manage wishlists, schedules, maps, and profile preferences.
- Provide an admin dashboard for managing users, activities, reports, and AI files.
- Integrate an AI/RAG chatbot that can answer questions using uploaded knowledge files.
- Use JWT-based authentication and role-based access to separate tourist and admin functionality.
- Deploy a modular architecture where frontend, backend, database, and AI service can evolve independently.

1.4 Project Scope

The implemented scope covers the tourist website, admin dashboard, ASP.NET Core backend, SQL Server persistence, image upload through Cloudinary, and a separate FastAPI RAG service. Refresh token and Remember Me behavior are intentionally excluded from the implemented feature documentation, and the current documented authentication flow is based on JWT access tokens and roles.

1.5 Methodology

1. Analyze functional requirements for tourist and admin users.
2. Design Angular standalone pages, services, guards, and interceptors.
3. Implement backend controllers, services, DTOs, and EF Core entities.
4. Implement AI/RAG file ingestion, chunking, vector storage, and chat retrieval.
5. Test APIs using Swagger/Postman-style flows and validate UI screens through browser testing.
6. Capture final UI screenshots and prepare project documentation.

Table 1: Technologies Used

Layer	Technology / Tool	Usage
Frontend	Angular 21 Standalone, Angular SSR	Main web application, routing, pages and hydration
Styling	Tailwind CSS, Flowbite, Font Awesome	Dark UI, components, icons and date pickers
Backend	ASP.NET Core 8 Web API	REST API, authentication, dashboard and business logic
Database	SQL Server / Azure SQL + EF Core	Persistent relational data and entity relationships
Images	Cloudinary	Profile and activity image hosting
AI/RAG	FastAPI, LangChain, OpenAI-compatible	Knowledge-based chatbot and document

NileGuide Graduation Project Documentation

	APIs	ingestion
Vector DB	Qdrant	Semantic vector search for retrieved chunks
Metadata DB	MongoDB	AI file and chunk metadata
Deployment	Azure App Service / Hosted Frontend / AI Host	Production execution environment

Chapter 2 - Background and Related Work

2.1 Tourism Planning Platforms

Modern tourism platforms commonly support discovery, filtering, details pages, favorites, scheduling, maps, and booking provider links. NileGuide follows this general model while focusing on Egypt activities and providing an admin-maintained content source.

2.2 AI Chatbots and RAG

Retrieval-Augmented Generation combines semantic search over a knowledge base with LLM answer generation. In NileGuide, uploaded documents are converted into chunks and embeddings. When a question is asked, the AI service retrieves relevant chunks and generates an answer using that context.

2.3 Research Gap

Many tourism websites provide static content but do not offer a maintainable AI knowledge base controlled from an admin interface. NileGuide fills this gap by connecting the tourist UI, admin dashboard, structured activity data, and RAG knowledge files in a single platform.

Chapter 3 - System Analysis and Requirements

3.1 Target Users

- Tourists who want to discover and plan activities in Egypt.
- Administrators who manage activities, users, reports, and AI knowledge files.
- Support or project team members who maintain tourism content and chatbot knowledge.

3.2 Functional Requirements

Table 2: Functional Requirements

Module	Requirement
Authentication	Register, login, forgot password, reset password and role-based access.
Activities	List, search, filter, sort, and view dynamic activity details.
Wishlist	Add/remove activities and view saved activities.
Schedule	Create a personal plan by scheduling selected activities.
Map	Display activity plan locations on an interactive map.
Profile	Update profile information, upload/delete profile picture, and manage preferences.
Admin Users	View, filter, update role/status, and delete users.
Admin Activities	Add, update, delete and list activities with images and metadata.
Reports	Display activity views, user growth, activities by category, and top activities.
AI/RAG	Upload, list, update, delete AI files and answer questions through chat.

3.3 Non-functional Requirements

- Responsive UI for common desktop resolutions.
- Secure protected routes using JWT bearer token and roles.
- Maintainable layered backend using controllers, services, DTOs, and entities.
- Scalable AI service separated from the main backend.
- Clear error handling and user feedback using forms, toasts and validation messages.

Chapter 4 - System Design and Architecture

4.1 System Architecture

NileGuide is designed as a modular web platform. Angular handles the presentation layer, ASP.NET Core manages structured application data, SQL Server stores relational records, and FastAPI handles RAG-specific AI tasks.

NileGuide System Architecture

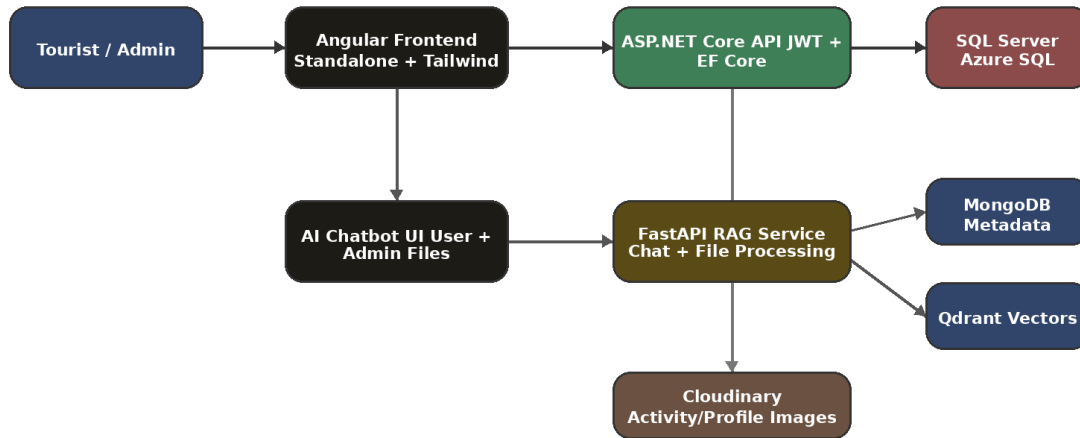


Figure 1: NileGuide system architecture

4.2 Frontend Architecture

The frontend uses Angular standalone routing, lazy loaded components, user and guest layouts, route guards, an authentication interceptor, and feature services. Components are organized by feature modules such as activities, details, wishlist, schedule, map, profile, contact, authentication, and admin dashboard.

Table 3: Main Frontend Routes

Route	Purpose	Access
/home	Landing page and public discovery entry	Public
/auth/login	User and admin login	Guest
/auth/register	Tourist registration	Guest
/activities	Activity listing with filters	Tourist
/activities/:id	Dynamic activity details page	Tourist
/wishlist	Saved activities	Tourist
/schedule	Personal itinerary	Tourist
/map	Interactive itinerary map	Tourist
/profile	Profile and preferences	Tourist
/contact	Contact form	Public
/dashboard/users-management	Admin user management	Admin
/dashboard/activities-management	Admin activity management	Admin
/dashboard/reports	Reports and analytics	Admin

Tourist User Flow

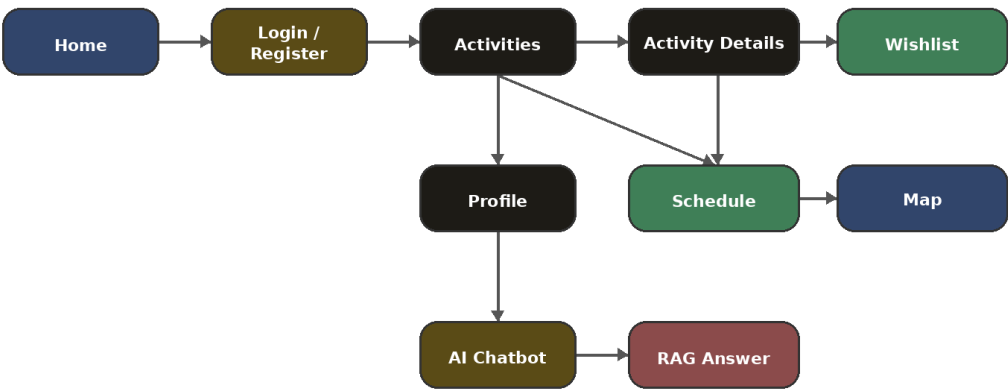


Figure 2: Tourist user flow

Admin Flow

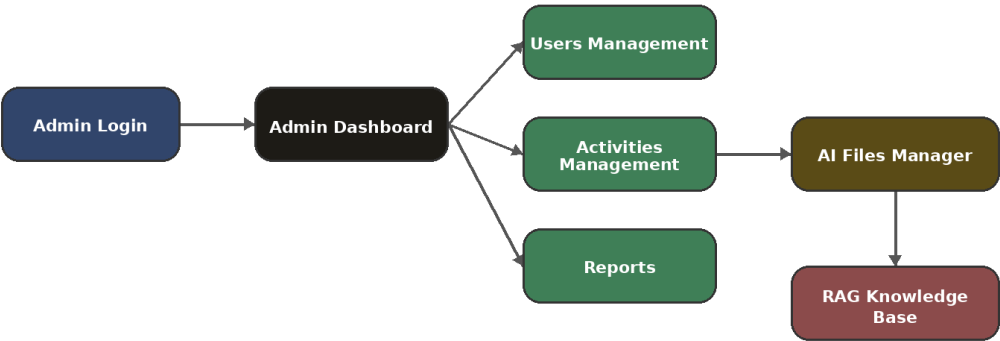


Figure 3: Admin flow

4.3 Backend Architecture

The backend is built using ASP.NET Core Web API. It registers EF Core, JWT authentication, CORS for the Angular frontend, rate limiting, Swagger, SignalR, Cloudinary profile picture services, email services, dashboard services, report services, and business services for activities, wishlist, plan, reviews and profiles.

Table 4: Main Backend Controllers

Controller	Main Responsibility
AuthController	Registration, login, current user, forgot password and reset password.
ActivitiesController	Public activity listing and activity details.
AdminActivitiesController	Admin CRUD operations for activities.
UsersController	Admin user management and profile picture endpoints.
UserProfilesController	Current user profile retrieval and update.
WishlistController	Wishlist items and status.
PlanController	Schedule/plan items and activity IDs.
ReviewsController	Activity reviews.
ReportsController	Activity views, user growth, category analytics and top activities.
DashboardController	Admin dashboard statistics.
CategoriesController / CitiesController	Lookup data for filters and profile preferences.
ChatConversationsController	Conversation ID ownership mapping for AI chat history.

JWT Authentication Sequence

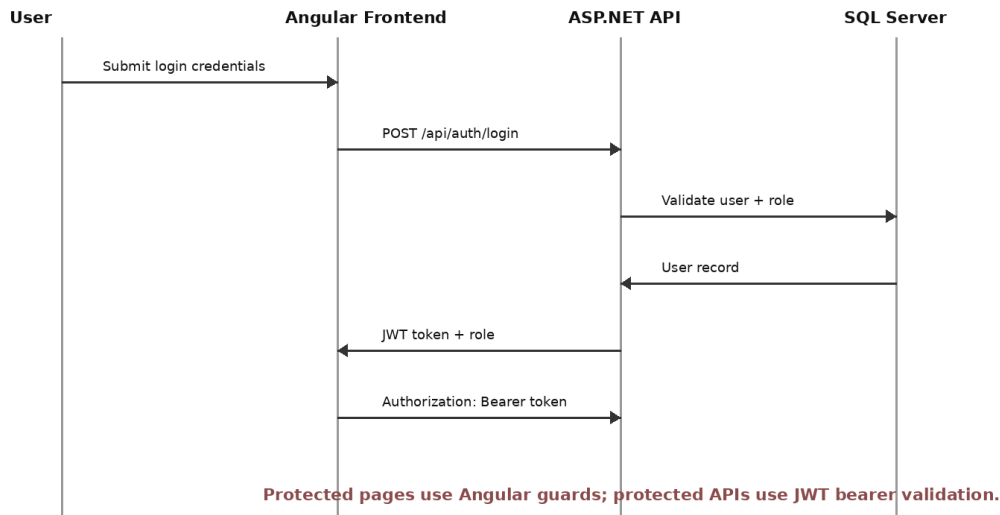
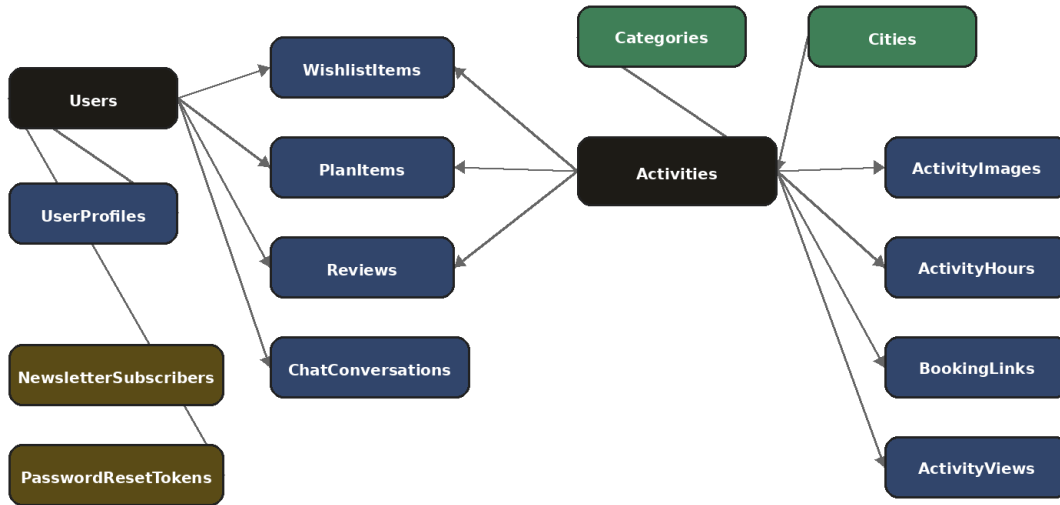


Figure 4: JWT authentication sequence

4.4 Database Design

The relational database is managed through Entity Framework Core. Users are linked to profiles, wishlist items, plan items, reviews and chat conversations. Activities are linked to categories, cities, images, opening hours, booking links, reviews, wishlist entries, plan entries and view records.

Core Database Relationships



Note: Refresh token entities exist in code but are excluded from the delivered feature scope according to project decision.

Figure 5: Core database entity relationships

Table 5: Core Database Entities

Entity	Purpose
Users	Authentication, role, active status, profile picture and base user data.
UserProfiles	Travel preferences, travel dates, preferred cities and interest categories.
Activities	Tourism activity content, pricing, duration, location and status.
Categories / Cities	Lookup entities used for filtering and classification.
ActivityImages / ActivityHours / BookingLinks	Supporting activity media, availability and booking provider data.
WishlistItems	Saved activities for a tourist.
PlanItems	Scheduled activities with date and start time.
Reviews	Activity ratings and comments.
ActivityViews	Analytics events for activity reports.
ChatConversations	Conversation IDs owned by users.
NewsletterSubscribers / PasswordResetTokens	Marketing and password reset support data.

4.5 AI/RAG Design

The AI service is separated from the .NET backend because RAG processing has different dependencies and scaling needs. FastAPI handles file upload, text extraction, chunking, embeddings, vector storage and chat answering. MongoDB stores file and chunk metadata, while Qdrant stores vectors for semantic retrieval.

RAG Upload and Chat Flow

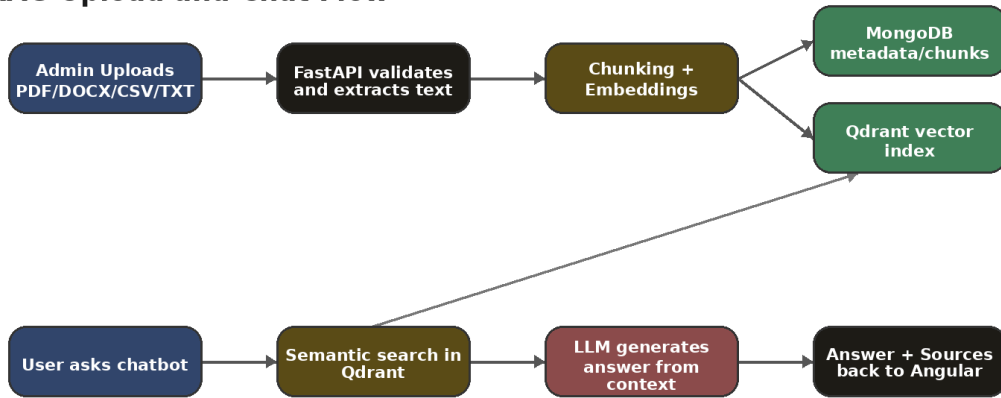


Figure 6: RAG upload and chat flow

Deployment View

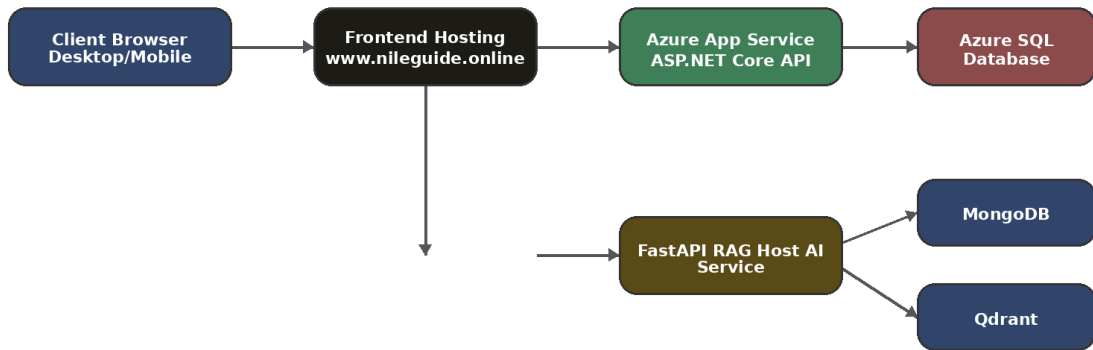


Figure 7: Deployment view

Chapter 5 - Implementation

5.1 Frontend Implementation

The Angular implementation uses feature-specific services to call backend APIs. ActivitiesService handles activity list, details, categories, cities and reviews. WishlistService handles saved activities. ScheduleService manages plan items. ProfileService manages current user profile and profile picture. Admin services support dashboard users, activities, reports and AI files.

- Angular guards separate guests, tourists and admins.
- The auth interceptor attaches the JWT token to protected backend calls.
- Flowbite date picker is used for date inputs where applicable.
- The UI uses a dark theme, yellow accent color and responsive Tailwind CSS layouts.
- The user chatbot is available from the user interface, while the admin AI file manager is available in the dashboard.

5.2 Backend Implementation

The ASP.NET Core backend is organized using controllers for HTTP endpoints, services for business logic, DTOs for request/response contracts, and EF Core models for persistence. Image upload uses multipart/form-data and Cloudinary storage. Validation errors are returned in a frontend-friendly JSON shape.

5.3 AI/RAG Implementation

The AI module exposes endpoints under /api/v1. Admin users upload documents to /data/multi-upload. Files are extracted, split into chunks, embedded and stored. Users submit chat messages to /nlp/chat and receive answers generated from retrieved knowledge context.

Table 6: AI/RAG Endpoints

Endpoint	Method	Purpose
/	GET	Health/base endpoint.
/api/v1/data/multi-upload	POST	Upload one or more knowledge files.
/api/v1/data/files	GET	List uploaded AI files.
/api/v1/data/file/{file_id}	PUT	Update an existing AI file.
/api/v1/data/file/{file_id}	DELETE	Delete an AI file and its related chunks/vectors.
/api/v1/nlp/chat	POST	Send a chat message and receive a RAG answer.

5.4 API Documentation

The backend API exposes authentication, activities, lookup data, wishlist, plan, profile, dashboard, reports, reviews, newsletter, and admin management endpoints. Appendix A provides a summarized API table for the endpoints used by the frontend.

Chapter 6 - Graphical User Interface

6.1 Tourist Screens

The tourist UI focuses on simple discovery and planning. The dark theme is consistent across public pages, user pages, and the admin area. The following screenshots are excerpts from the final web application.

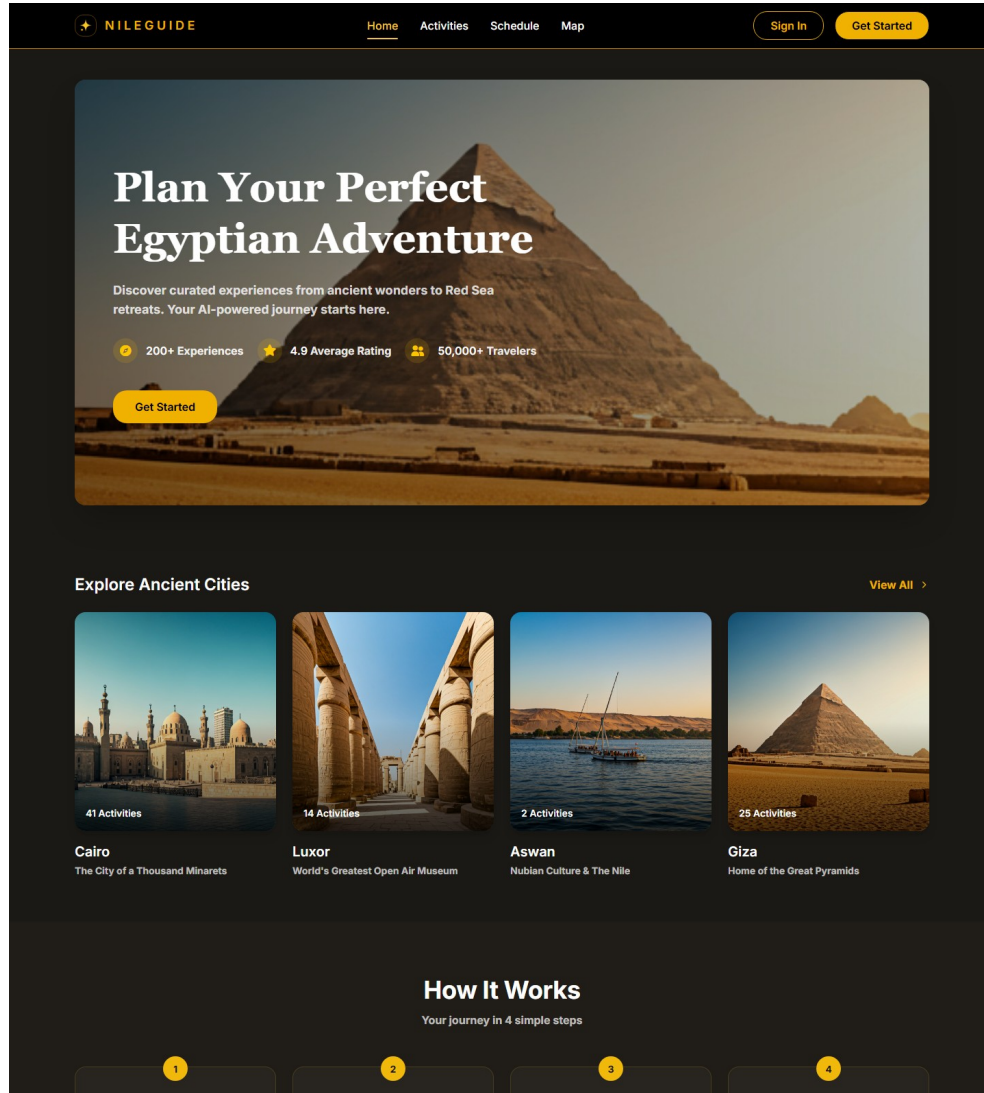


Figure 8: Home page

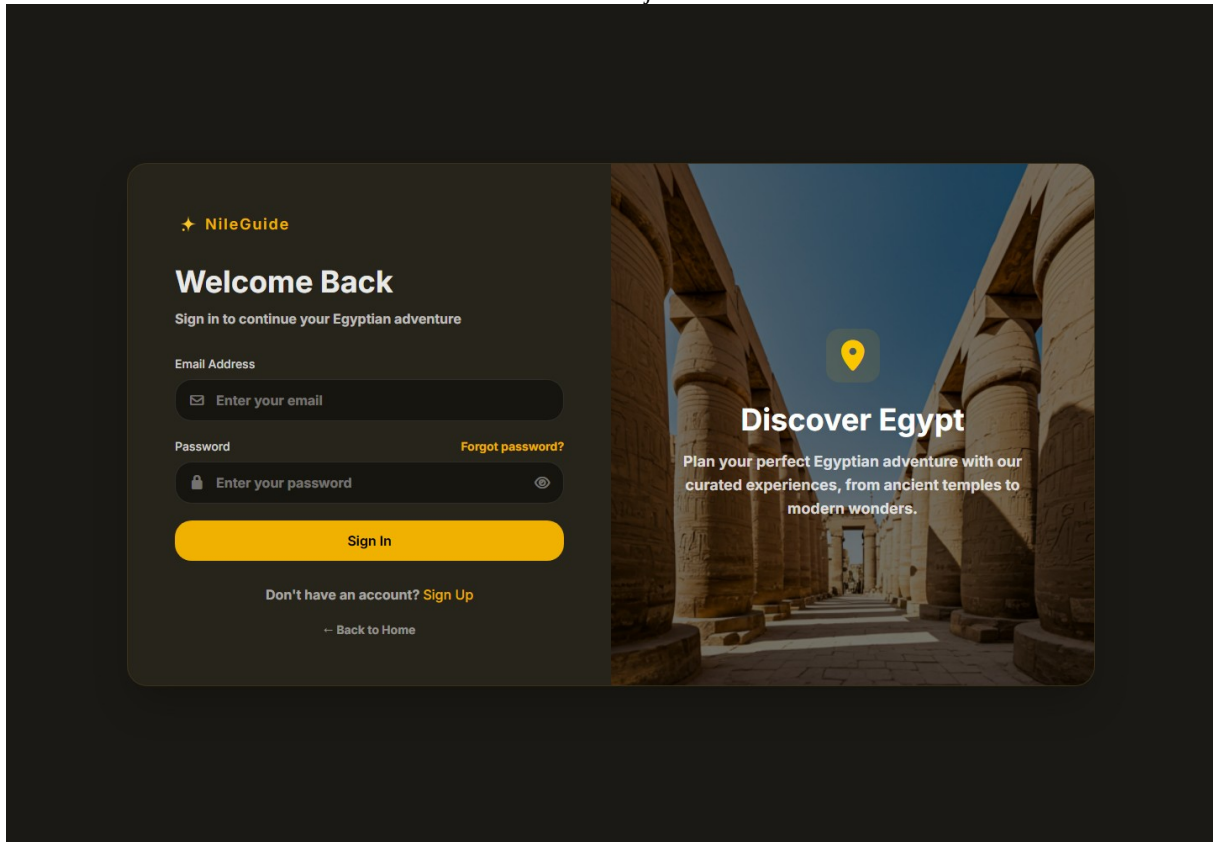


Figure 9: Login page

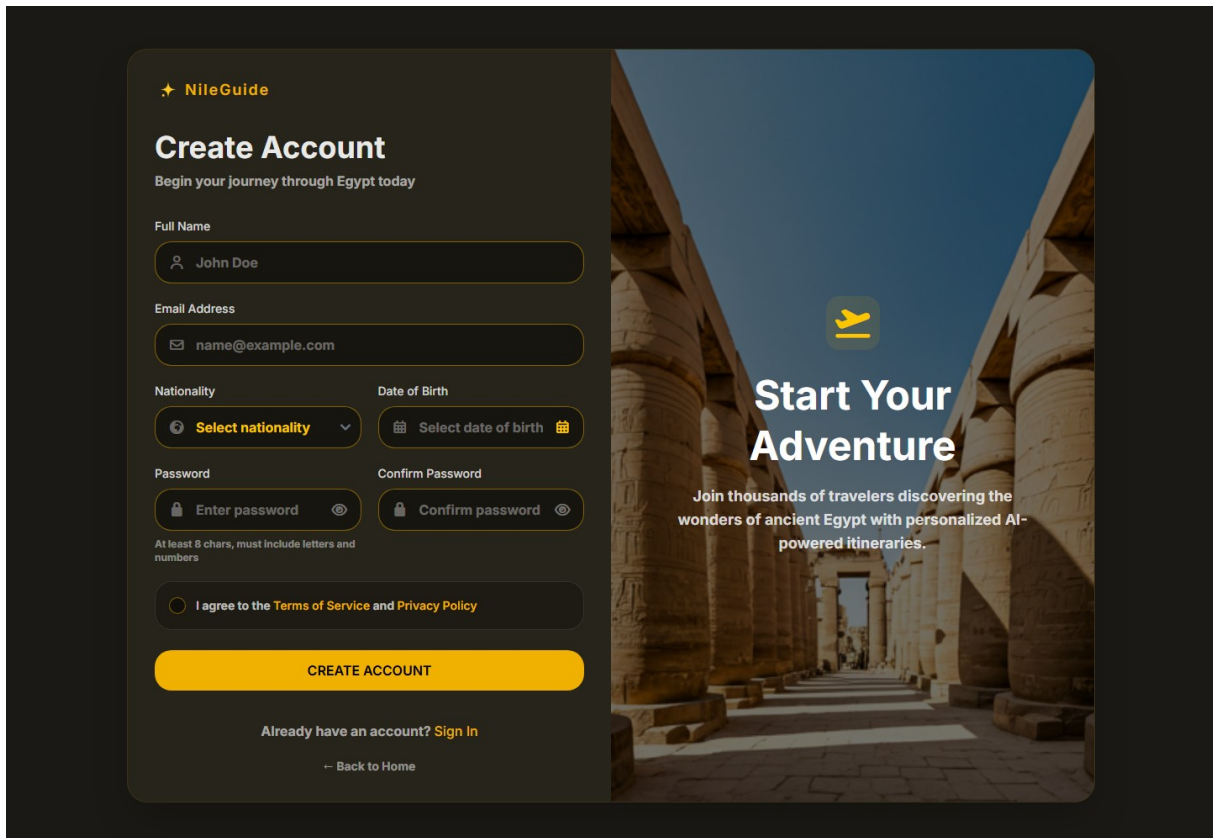


Figure 10: Registration page

[Home](#)
[Activities](#)
[Schedule](#)
[Map](#)

Curated Experiences in Egypt

Categories

- ☐ Adventure & Safari
- ☐ Beaches & Islands
- ☐ Desert Trips
- ☐ Family Activities
- ☐ History & Culture
- ☐ Local Food
- ☐ Luxury & Relaxation
- ☐ Museums & Galleries
- ☐ Nightlife & Entertainment
- ☐ Nile Cruises
- ☐ Pyramids & Archaeology
- ☐ Red Sea & Diving
- ☐ Religious Landmarks
- ☐ Shopping & Souks
- ☐ Wellness & Spa

Cities

Alexandria

Assut

Aswan

Behelra

Beni Suet

Cairo

Dakahilla

Damietta

Fayoum

Gharbia

Giza

Ismailia

Kafr El Sheikh

Luxor

Matrouh

Menoufia

Minya

New Valley

North Sinal

Port Said

Qalyubia

Qena

Red Sea

Shangja

Soahg

South Sinal

Showing 104 Activities

Sort by: Popularity

ACTIVE

PYRAMIDS & ARCHAEOLOGY

★ 4.5 (2)

Giza Pyramids,Sphinx,Camel Ride,ATV Bike,Shopping, Dinn...

Mobile or paper ticket accepted; Adult required for booking

Talk about action-packed. This tour offers a variety of activities in one full...

OPEN 5:00 PM

CLOSE 11:00 PM

\$31.5

Book via:

Add To Plan

View Details

ACTIVE

PYRAMIDS & ARCHAEOLOGY

★ 5.0 (1242)

Pyramids Sphinx Camel ATV Bike Shopping and Nile Dinner...

Mobile or paper ticket accepted; Adult required for booking

Set on the outskirts of Cairo, the Giza Pyramids take time to reach by public...

OPEN 5:00 PM

CLOSE 11:00 PM

\$32

Book via:

Add To Plan

View Details

ACTIVE

NILE CRUISES

★ 5.0 (1083)

Luxury Felucca on the Nile with Lunch

Mobile or paper ticket accepted; Adult required for booking

Luxury sailing felucca on the Nile River there are many felucca ride on the nil...

OPEN 9:00 AM

CLOSE 11:00 AM

\$55

Book via:

Add To Plan

View Details

ACTIVE

FAMILY ACTIVITIES

★ 5.0 (1)

VIP Meet and Greet Private Hotel Transfer from Cairo Airport

Mobile or paper ticket accepted; Adult required for booking

Our target is to meet your expectations, for all those passengers who need to...

OPEN 12:00 AM

CLOSE 11:00 PM

\$39

Book via:

Add To Plan

View Details

ACTIVE

HISTORY & CULTURE

★ 4.8 (4)

Hot Air Balloon Ride in Luxor Egypt with Transfers Included

Mobile or paper ticket accepted

Safety is our #1 priority for passengers and crewmembers, with over 30 year...

OPEN 4:00 AM

CLOSE 7:00 AM

\$85

Book via:

Add To Plan

View Details

ACTIVE

PYRAMIDS & ARCHAEOLOGY

★ 4.9 (1296)

All inclusive Giza Pyramids,Sphinx,lunch,Camel,...

Mobile or paper ticket accepted

The tour is VIP service with no hassle or scam .as soon as you book the tour...

OPEN 8:00 AM

CLOSE 1:00 PM

\$8

Book via:

Add To Plan

View Details

Figure 11: Activities listing page

20

← Back

PYRAMIDS & ARCHAEOLOGY
Verified

Giza Pyramids, Sphinx, Camel Ride, ATV Bike, Shopping, Dinner cruise

★★★★★ 4.5 (2 reviews)

6h

Giza

\$31.5

Save

Share

DURATION

6h

GROUP SIZE

Up to 8 travelers

PRICE BASIS

per person

CANCELLATION

For a full refund, cancel at least 24 hours before the scheduled departure time.

29.977° N - 31.333° E

Open Google Maps

Open in Google Maps

Meeting Point

Giza

Get Directions on Google Maps

About this Activity

Talk about action-packed. This tour offers a variety of activities in one full day of adventures. First, visit the Giza pyramid complex and the Great Sphinx, then enjoy your ATV ride and camel ride in the surrounding desert. After a break to shop for souvenirs, it's time to get ready for a dinner cruise and belly dancing show on the Nile River. So what are you waiting for? Book now and enjoy this one-of-a-kind experience.

Gallery

View All Photos





Required Documents

Please prepare these documents before arrival.

Mobile or paper ticket accepted; Adult required for booking

MANDATORY

Reviews

2 traveler reviews for Giza Pyramids, Sphinx, Camel Ride, ATV Bike, Shopping, Dinner cruise.

ahmed

London · Jun 15, 2025

★★★★★

4.5 TRAVELER RATING

Add Your Review

Share your experience and let other travelers know what you think.

Your City

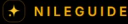
5 Stars

Write your review

Submit Review

Figure 12: Activity details page

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NILEGUIDE

HomeActivitiesScheduleMap

AE

My Wishlist
Activities you want to experience



2 Activities Saved



ACTIVE

★ 5.0 (1242)

Pyramids Sphinx Camel ATV Bike Shopping and Nile Dinner Cruise

Mobile or paper ticket accepted; Adult required for booking

Set on the outskirts of Cairo, the Giza Pyramids take time to reach by public transport. This grea...

OPEN5:00 PM

CLOSE11:00 PM

\$32

Book via:   

Add To Plan

View Details



ACTIVE

★ 4.5 (2)

Giza Pyramids, Sphinx, Camel Ride, ATV Bike, Shopping, Dinner cruise

Mobile or paper ticket accepted; Adult required for booking

Talk about action-packed. This tour offers a variety of activities in one full day of adventures...

OPEN5:00 PM

CLOSE11:00 PM

\$31.5

Book via:   

Add To Plan

View Details

NileGuide

Your premium AI companion for exploring the wonders of ancient and modern Egypt.

Cairo, Egypt

hello@nileguide.com

Explore

Home

Activities

My Schedule

Map

Account

Profile

Wishlist

Sign In

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Figure 13: Wishlist page

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[Home](#)
[Activities](#)
[Schedule](#)
[Map](#)

My Egypt Schedule

Export

Total Activities
2

Total Duration
13h

Total Cost
\$88

TIME	ACTIVITY	LOCATION	OPEN	CLOSE	DURATION	COST	ACTIONS
Jun 25, 2026 9:00 AM to 1:00 PM	All inclusive Giza Pyramids, Sphinx, lunch, Camel, insid pyramid The tour is VIP service with no hassle or scam .as soon as you book the tour, our...	Giza	8:00 AM	1:00 PM	4h	\$8	
Jun 25, 2026 1:30 PM to 10:30 PM	Private Day Tour To El Fayoum From Cairo Private Day Tour To El Fayoum From Cairo Escape the hustle and bustle of Cairo for...	Cairo	8:00 AM	5:00 PM	9h	\$80	

+ Add New Activity

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📍 Cairo, Egypt

✉️ hello@nileguide.com

Explore

Home

Activities

My Schedule

Map

Account

Profile

Wishlist

Sign In

Support

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Figure 14: Schedule page

[Home](#)
[Activities](#)
[Schedule](#)
[Map](#)

Interactive Map

All Cities

Itinerary
2 Items

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📍 Cairo, Egypt

✉️ hello@nileguide.com

Explore

Home

Activities

My Schedule

Map

Account

Profile

Wishlist

Sign In

Support

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Figure 15: Interactive map page

 NILEGUIDE

HomeActivitiesScheduleMap

  AE

AT

Welcome to Egypt.
Let's tailor your journey.

Tell us about yourself so our AI can craft the perfect itinerary, blending ancient wonders with modern comfort.

Basic Information

Name

Abdelrahman tamer emam

Email

botta@gmail.com

Date of Birth

2000-01-01

Where are you visiting from?

Andorran

Do you have travel dates?

Add your start and end dates if your trip is already planned.

+ Add travel dates

Preferred Cities

☐ Alexandria

☐ Assiut

☐ Aswan

☐ Beheira

☐ Beni Suef

☐ Cairo

☐ Dakahlia

☐ Damietta

☐ Fayoum

☐ Gharbia

☐ Giza

☐ Ismailia

☐ Kafr El Sheikh

☐ Luxor

☐ Matrouh

☐ Menoufia

☐ Minya

☐ New Valley

☐ North Sinai

☐ Port Said

☐ Qalyubia

☐ Qena

☐ Red Sea

☐ Sharqia

☐ Sohag

☐ South Sinai

☐ Suez

What interests you most?

Adventure & Safari

Beaches & Islands

Desert Trips

Family Activities

History & Culture

Local Food

Luxury & Relaxation

Museums & Galleries

Nightlife & Entertainment

Nile Cruises

Pyramids & Archaeology

Red Sea & Diving

Religious Landmarks

Shopping & Souks

Wellness & Spa

Save My Profile →

Figure 16: Profile page

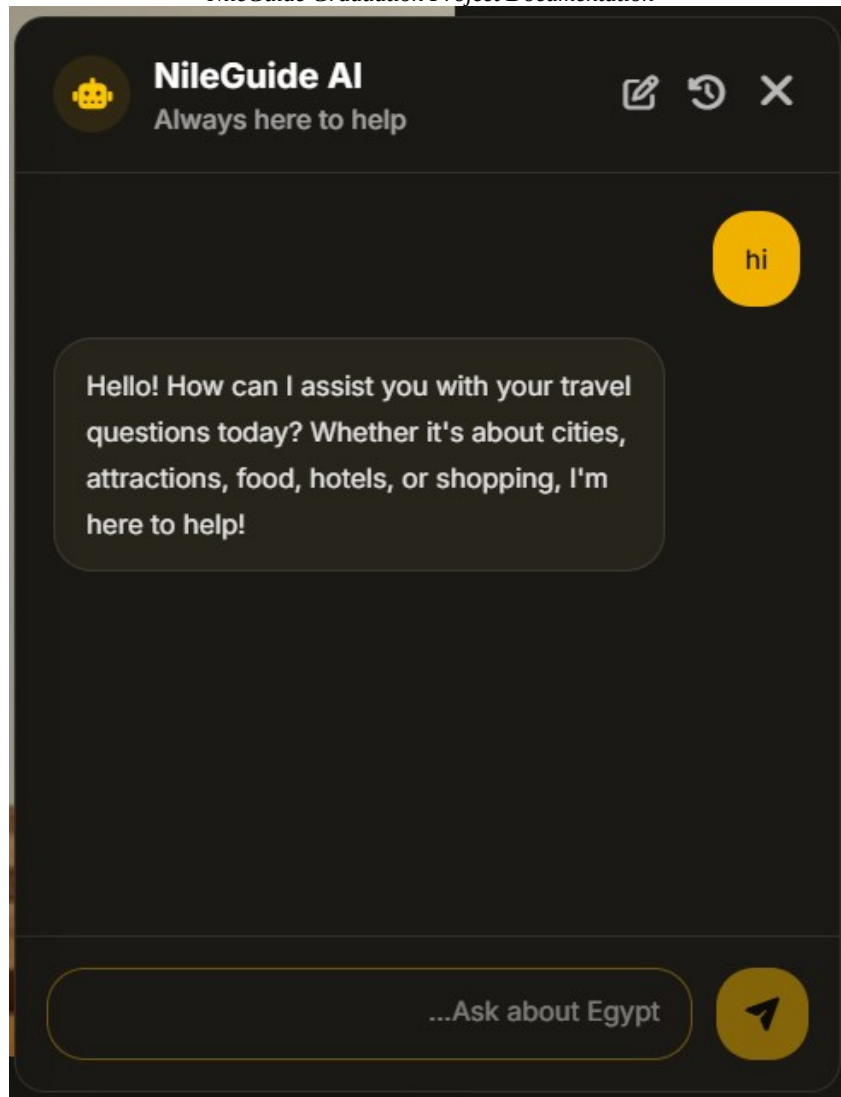


Figure 17: User chatbot panel

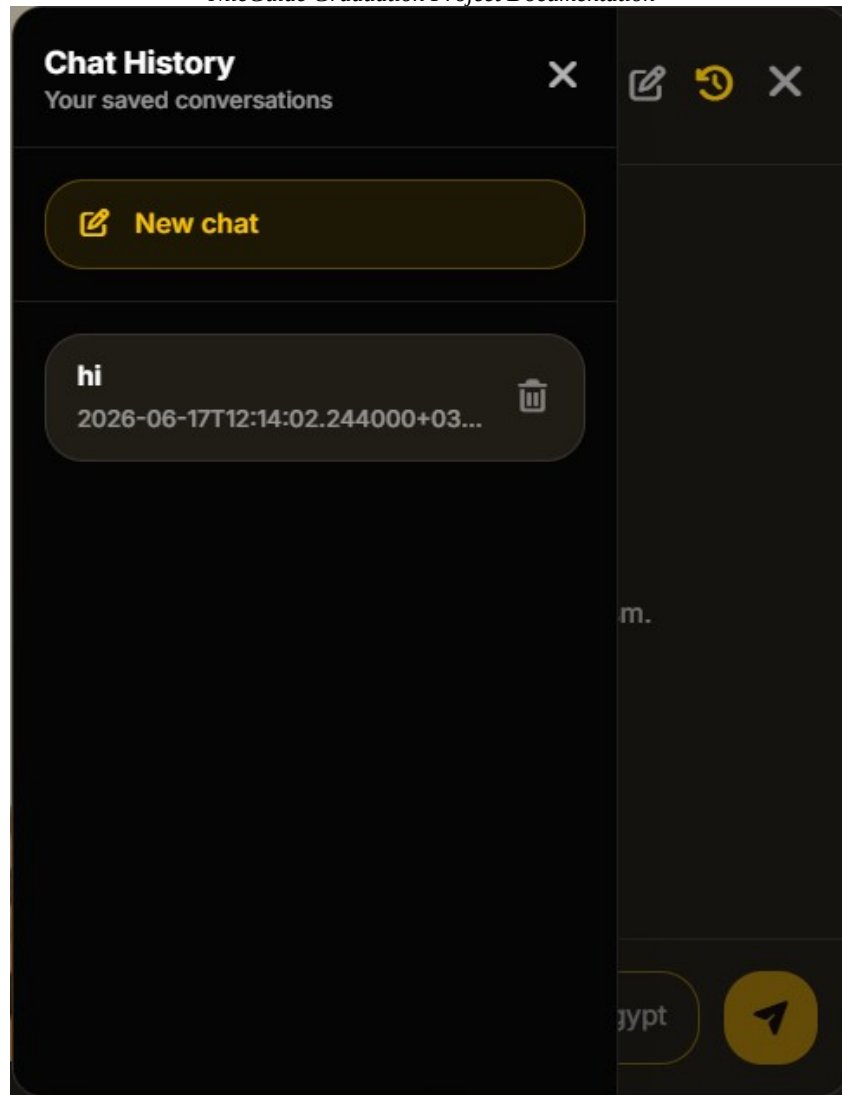


Figure 18: User chatbot history panel

 NILEGUIDE

HomeActivitiesScheduleMap

Sign InGet Started



Get in Touch

Have a question or need assistance? We are here to help you plan your perfect Egyptian adventure.

Send Us a Message

Fill in the form below and we will get back to you as soon as possible.

Full Name

John Doe

Email Address

john@example.com

Subject

Select a subject

Message

Tell us how we can help...

Send Message



Response Time

We typically respond within 2 hours during business hours.



Email Us

General Support
nileguide.noreply@gmail.com

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Cairo, Egypt

hello@nileguide.com

Explore

HomeActivitiesMy ScheduleMap

Account

ProfileWishlistSign In

Support

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Figure 19: Contact page

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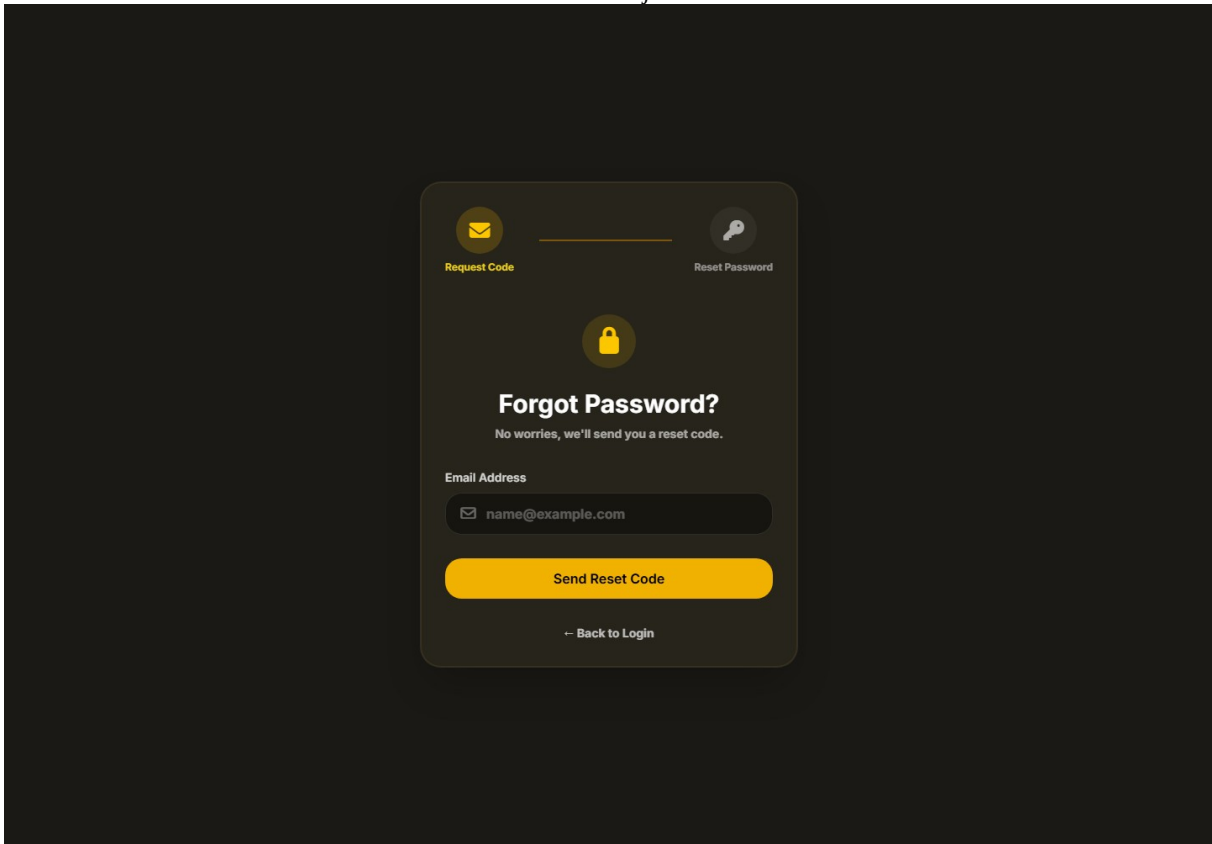


Figure 20: Forgot password page

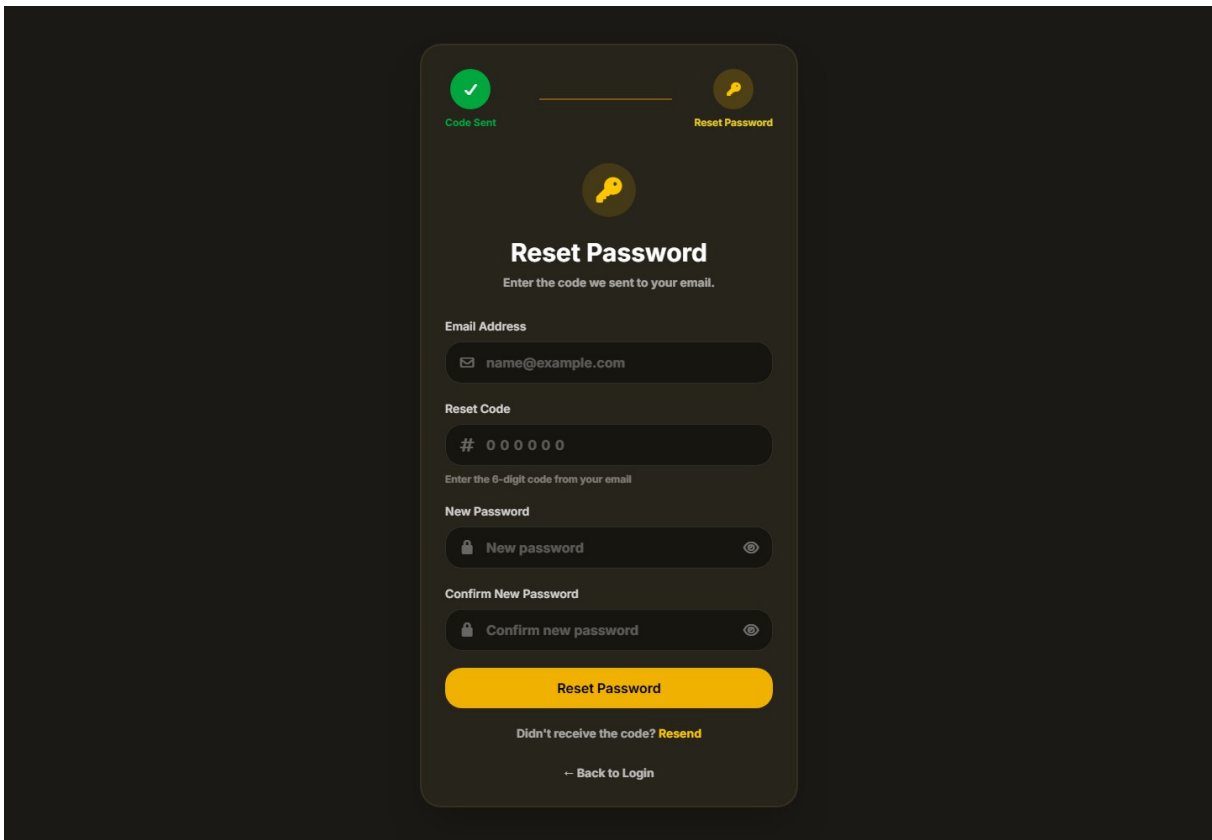


Figure 21: Reset password page

6.2 Admin Screens

The admin dashboard provides management screens for users, activities, reports and AI files. The dashboard route redirects to activities management by default in the Angular routing configuration, so the available admin screenshots focus on the implemented management screens.

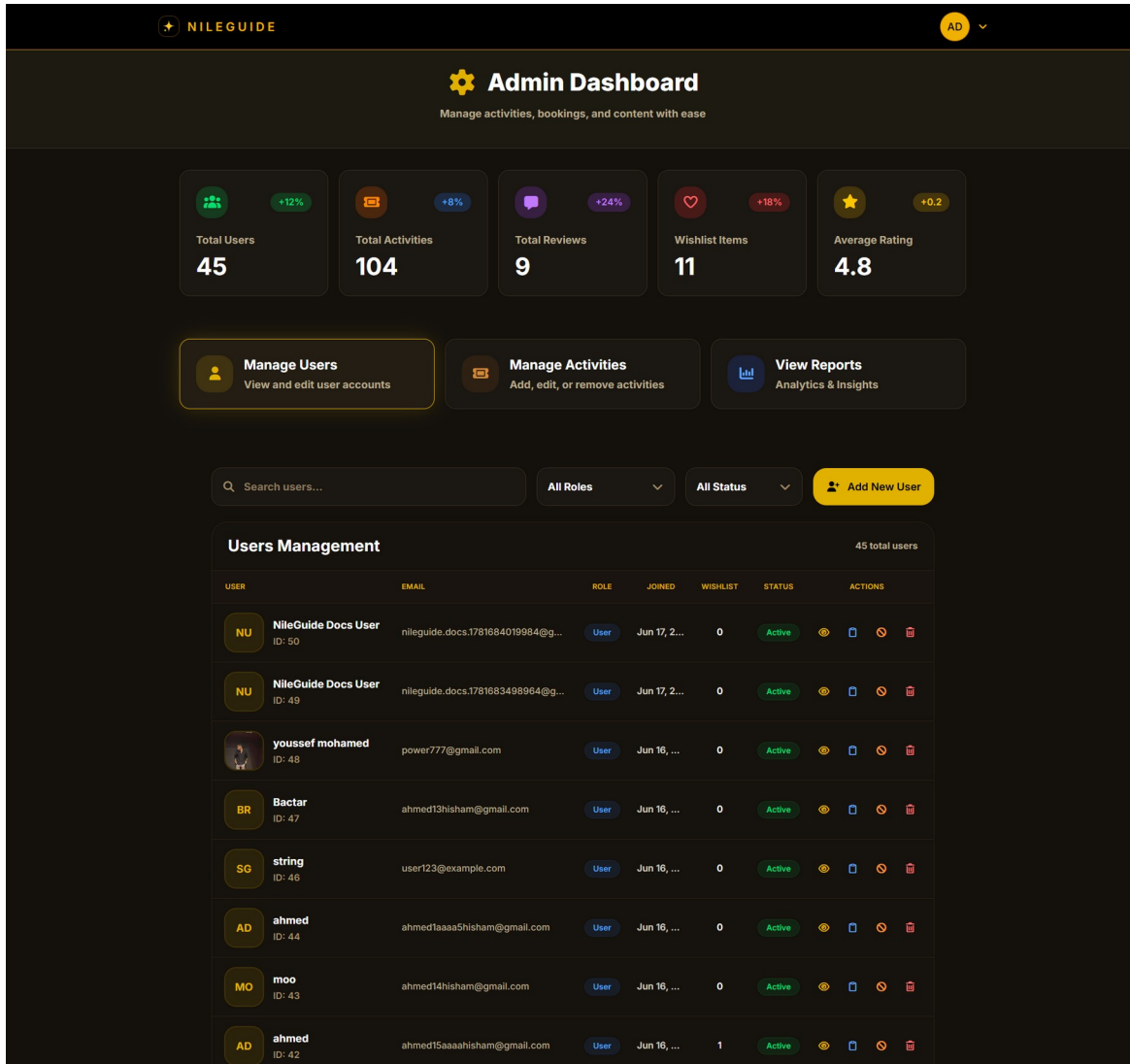


Figure 22: Admin users management screen

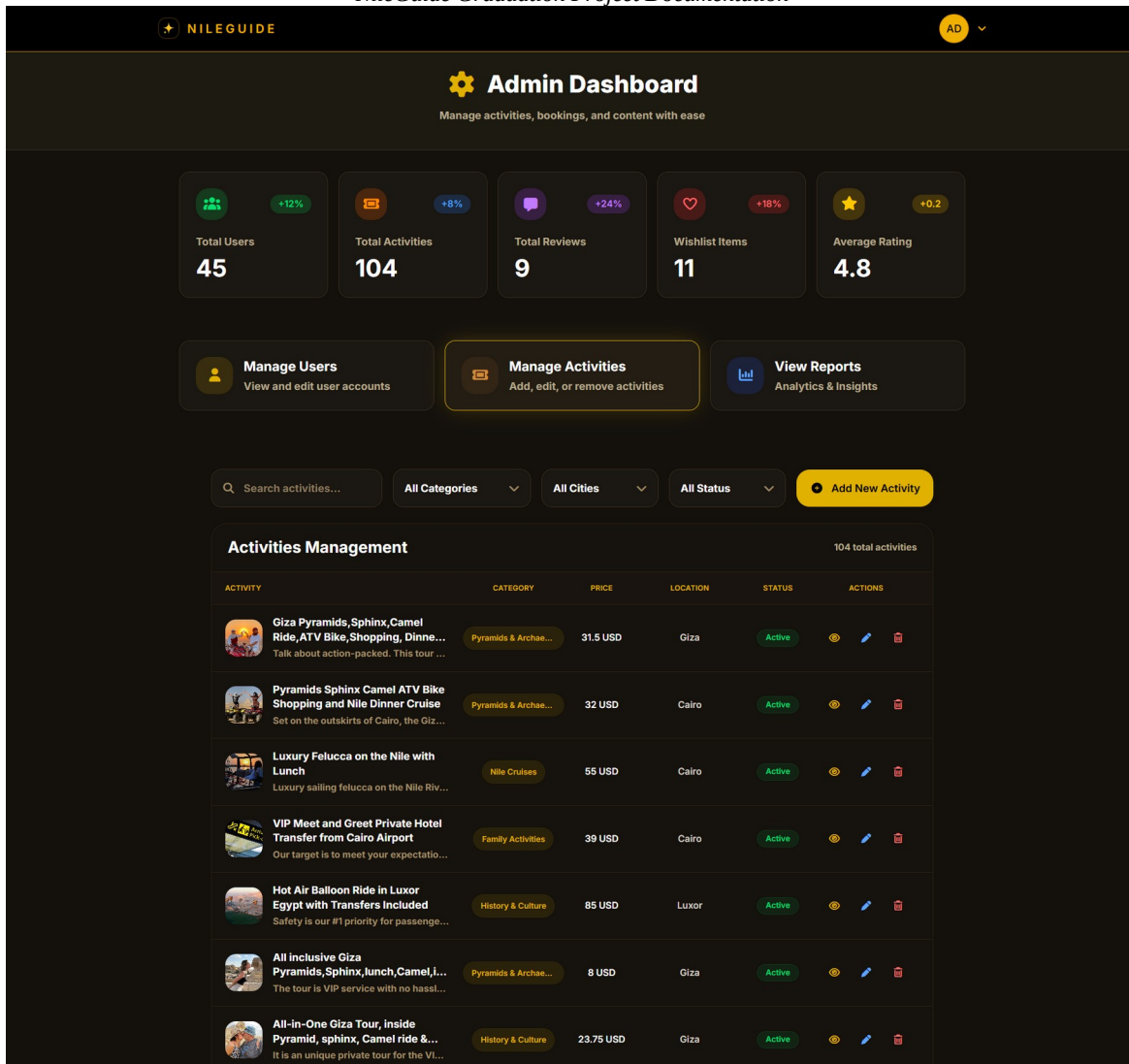


Figure 23: Admin activities management screen

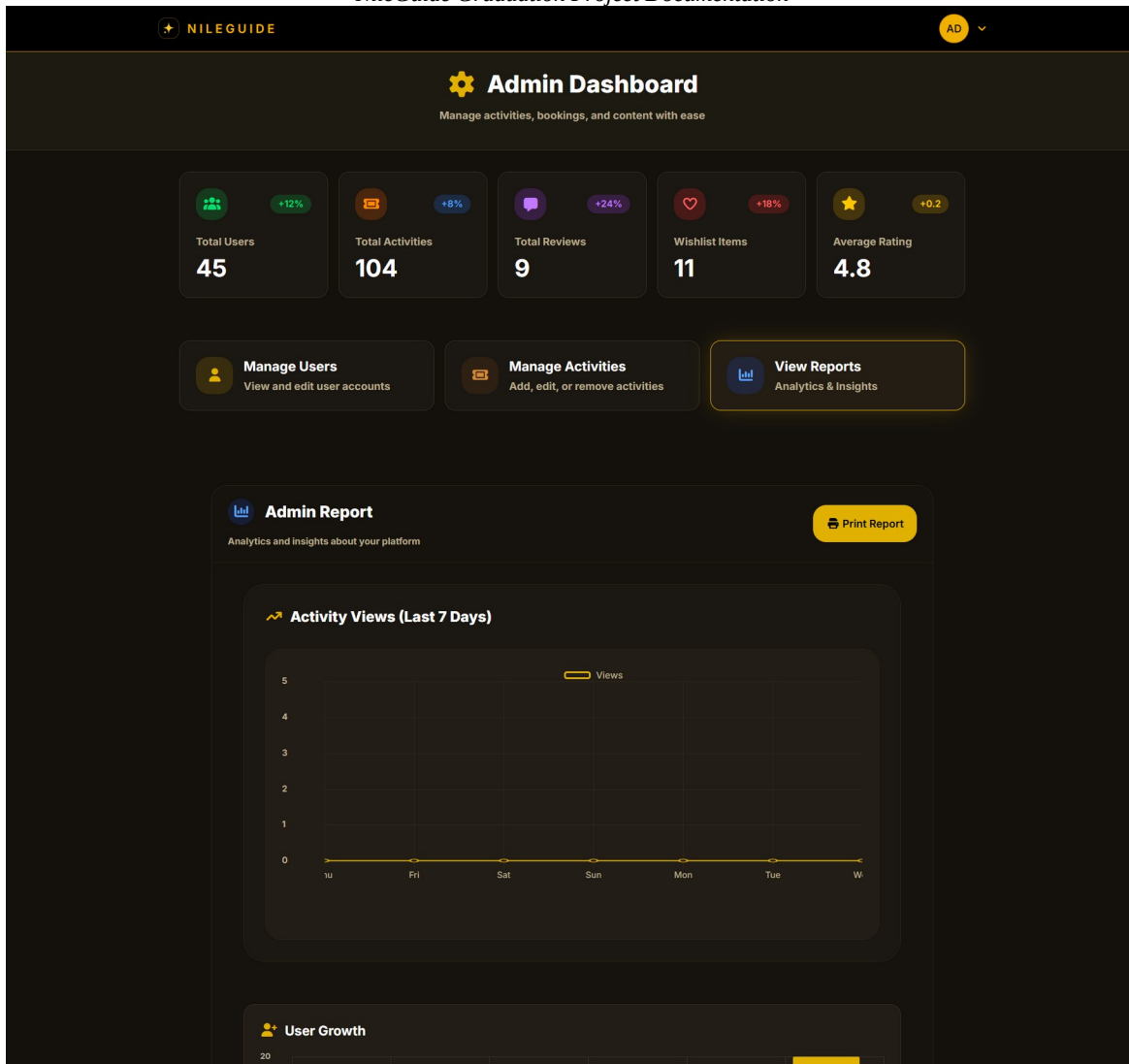


Figure 24: Admin reports screen

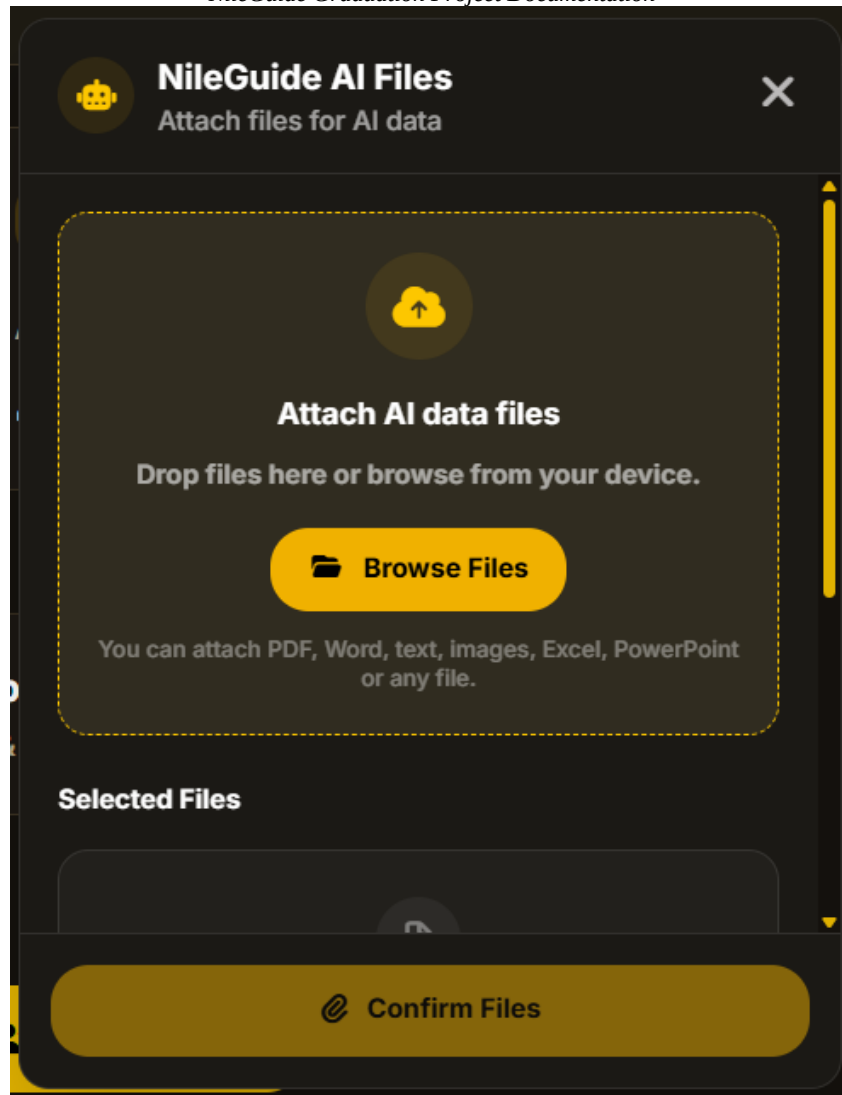


Figure 25: Admin AI files manager

Chapter 7 - Testing, Challenges and Deployment

7.1 Testing Scope

Table 7: Testing Scope

Area	Tested Scenario
Authentication	Register, login, forgot password and reset password flows.
Activities	List loading, filters, cards, details route and images.
Wishlist	Add/remove saved activities and fetch activity IDs.
Schedule	Create and display scheduled activities.
Profile	Load/update profile data and profile picture.
Admin	Users, activities and reports screens.
AI/RAG	File upload/list/update/delete and chat response endpoint.
Build	Angular build and backend API execution/deployment checks.

7.2 Challenges and Solutions

Table 8: Challenges and Solutions

Challenge	Solution
Protected route access	Angular guards and JWT bearer API validation separate tourist and admin screens.
SSR/localStorage issues	Browser checks and safe token access prevent server-side localStorage errors.
Activity images	Multipart/form-data upload is used and image URLs are stored after Cloudinary upload.
AI knowledge updates	Admin AI files manager allows updating the RAG knowledge base without code changes.
Profile avatar fallback	The UI displays uploaded profile pictures or initials when no image exists.
Report generation	A dedicated ReportsService fetches activity views, user growth, category distribution and top activity data.

7.3 Deployment

The documented deployment model uses a hosted Angular frontend, an ASP.NET Core API hosted on Azure App Service, Azure SQL for relational data, Cloudinary for images, and a separately hosted FastAPI RAG service connected to MongoDB and Qdrant. The live frontend URL used by the project is <https://www.nileguide.online>.

Chapter 8 - Conclusion and Future Work

8.1 Conclusion

NileGuide delivers a complete tourism planning experience by connecting discovery, planning, administration and AI assistance in a modular system. The Angular frontend provides the interface, the ASP.NET Core backend manages trusted structured data, and the FastAPI RAG service extends the platform with knowledge-based chat support.

8.2 Future Work

Table 9: Future Work

Future Feature	Description
Booking integration	Connect real booking/payment providers instead of only external provider links.
Advanced recommendations	Use user preferences and behavior to recommend personalized activities.
Notifications	Notify users about planned activities or admin updates.
Multi-language support	Add Arabic/English language switching across the full interface.
Mobile application	Provide a native or PWA experience for tourists.
Extended AI history	Improve long-term saved AI conversation history and feedback analytics.
Refresh tokens / Remember Me	Can be added later as a secure session enhancement, but not documented as implemented in the current scope.

References

- [1] Angular Documentation, Angular Standalone Components and Routing.
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- [4] FastAPI Documentation, Modern Python API Development.
- [5] Qdrant Documentation, Vector Database and Similarity Search.
- [6] MongoDB Documentation, Document Database Concepts.
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- [8] Tailwind CSS Documentation, Utility-First Styling.
- [9] Flowbite Documentation, UI Components for Tailwind CSS.

Appendix A - API Summary

Table 10: API Summary

Method	Endpoint	Purpose
POST	/api/auth/register	Create a tourist account.
POST	/api/auth/login	Authenticate and return JWT access token and role.
GET	/api/auth/me	Return current authenticated user.
POST	/api/auth/forgot-password	Send password reset code.
POST	/api/auth/verify-reset-code	Verify reset code.
POST	/api/auth/reset-password	Reset account password.
GET	/api/Activities	List activities with filters/pagination.
GET	/api/Activities/{id}	Return details for one activity.
GET	/api/Categories	List categories.
GET	/api/Cities	List cities.
GET	/api/activities/{activityId}/reviews	List reviews for an activity.
POST	/api/activities/{activityId}/reviews	Create an activity review.
GET	/api/wishlist	Get current user wishlist.
GET	/api/wishlist/activity-ids	Get saved activity IDs.
POST	/api/wishlist/{activityId}	Add activity to wishlist.
DELETE	/api/wishlist/{activityId}	Remove activity from wishlist.
GET	/api/wishlist/status/{activityId}	Check if activity is saved.
GET	/api/plan	Get user schedule.
POST	/api/plan/items	Add activity to schedule.
DELETE	/api/plan/items/{planItemId}	Delete schedule item.
GET	/api/plan/activity-ids	Get planned activity IDs.
GET	/api/users/me/profile	Get current user profile.
PUT	/api/users/me/profile	Update current user profile.
POST	/api/users/me/profile-picture	Upload profile picture.
DELETE	/api/users/me/profile-picture	Delete profile picture.
GET	/api/users	Admin list users.
GET	/api/users/{id}	Admin get user details.
POST	/api/users	Admin create user.
PATCH	/api/users/{id}/role	Admin update user role.
PATCH	/api/users/{id}/status	Admin update user status.
DELETE	/api/users/{id}	Admin delete user.
GET	/api/admin/activities	Admin list activities.
POST	/api/admin/activities	Admin create activity using multipart/form-data.
PUT	/api/admin/activities/{id}	Admin update activity using multipart/form-data.
DELETE	/api/admin/activities/{id}	Admin delete activity.
GET	/api/dashboard	Admin dashboard statistics.
GET	/api/reports/activity-views	Activity views report.
GET	/api/reports/user-growth	User growth report.
GET	/api/reports/activities-by-category	Activities by category report.
GET	/api/reports/top-activities	Top activities report.
POST	/api/v1/data/multi-upload	AI/RAG upload files.
GET	/api/v1/data/files	AI/RAG list files.
PUT	/api/v1/data/file/{file_id}	AI/RAG update file.
DELETE	/api/v1/data/file/{file_id}	AI/RAG delete file.
POST	/api/v1/nlp/chat	AI/RAG chat endpoint.